

Digital Nurture 5.0 Assessment

50-Question Compact Practice Set

I. Design Patterns & SOLID

- Q1.** Which SOLID principle states that a class should have only one reason to change?
- A) Open/Closed Principle
 - B) Single Responsibility Principle
 - C) Liskov Substitution Principle
 - D) Interface Segregation Principle
- Q2.** In Spring, which design pattern is primarily used by the BeanFactory?
- A) Factory Pattern
 - B) Singleton Pattern
 - C) Decorator Pattern
 - D) Observer Pattern
- Q3.** By default, Spring beans are created using which design pattern?
- A) Prototype
 - B) Builder
 - C) Singleton
 - D) Strategy
- Q4.** Which pattern is used to add new functionality to an existing object without altering its structure?
- A) Facade
 - B) Adapter
 - C) Decorator
 - D) Proxy
- Q5.** Spring AOP (Aspect-Oriented Programming) heavily relies on which structural pattern?
- A) Proxy Pattern
 - B) Bridge Pattern
 - C) Flyweight Pattern
 - D) Composite Pattern
- Q6.** Which pattern defines a family of algorithms, encapsulates each one, and makes them interchangeable?
- A) State Pattern
 - B) Command Pattern
 - C) Strategy Pattern
 - D) Template Method Pattern

- Q7.** The Liskov Substitution Principle (LSP) ensures that:
- A)** Interfaces should be small and specific
 - B)** Derived classes must be substitutable for their base classes
 - C)** High-level modules should not depend on low-level modules
 - D)** Classes should be open for extension but closed for modification
- Q8.** Which behavioral pattern is used by React's context or Redux store to notify components of state changes?
- A)** Iterator
 - B)** Mediator
 - C)** Observer
 - D)** Memento
- Q9.** What design pattern provides a simplified interface to a larger body of code, such as a class library?
- A)** Facade Pattern
 - B)** Adapter Pattern
 - C)** Proxy Pattern
 - D)** Bridge Pattern
- Q10.** Dependency Injection (DI) is a specific implementation of which broader principle?
- A)** Interface Segregation
 - B)** Inversion of Control (IoC)
 - C)** Open/Closed Principle
 - D)** Encapsulation

II. Spring, Maven & Hibernate

- Q11.** In Hibernate, what is the 'N+1 selects' problem?
- A)** Saving N entities requires N+1 database connections
 - B)** Fetching 1 parent entity and its N children results in N+1 database queries
 - C)** The L1 cache allows only N+1 objects
 - D)** A pagination offset issue
- Q12.** Which Hibernate method is used to re-attach a detached object to the current session?
- A)** session.save()
 - B)** session.update()
 - C)** session.persist()
 - D)** session.load()
- Q13.** What is the primary difference between session.save() and session.persist() in Hibernate?
- A)** save() returns the generated identifier; persist() returns void
 - B)** persist() returns the generated identifier; save() returns void
 - C)** There is no difference

D) persist() only works with JPA annotations

Q14. In Maven, which phase compiles the source code of the project?

- A) validate
- B) compile
- C) test
- D) package

Q15. What does the <scope>test</scope> tag in a Maven dependency mean?

- A) The dependency is included in the final artifact
- B) The dependency is used for compilation only
- C) The dependency is only required for compiling and running tests
- D) The dependency is provided by the application server

Q16. Which cache is enabled by default in Hibernate?

- A) First-level cache (L1)
- B) Second-level cache (L2)
- C) Query cache
- D) None

Q17. How do you specify a composite primary key in JPA/Hibernate?

- A) @CompositeKey
- B) @IdClass or @EmbeddedId
- C) @PrimaryKeyColumn
- D) @Multid

Q18. In Spring, which annotation is used to inject dependencies automatically by type?

- A) @Resource
- B) @Inject
- C) @Autowired
- D) @Value

Q19. What is a major advantage of using HikariCP in a Spring Boot application?

- A) It replaces Hibernate as an ORM
- B) It is a high-performance JDBC connection pool
- C) It handles JSON serialization
- D) It manages Spring Security filters

Q20. What is the correct way to map a One-to-Many relationship in Hibernate without creating a separate join table?

- A) Use @JoinTable on the parent
- B) Use @JoinColumn on the child and mappedBy on the parent
- C) Use @ManyToMany

D) Only use @OneToMany without any other annotations

III. Spring Boot & REST

Q21. Which dependency is required to expose operational endpoints like /health and /metrics in Spring Boot?

- A) spring-boot-starter-web
- B) spring-boot-starter-actuator
- C) springboot-starter-security
- D) spring-boot-admin-starter

Q22. Which annotation allows Spring Boot to configure beans based on classpath settings, other beans, and property settings?

- A) @ComponentScan
- B) @Configuration
- C) @EnableAutoConfiguration
- D) @SpringBootApplication

Q23. If you want a REST controller to return JSON automatically without using @ResponseBody on every method, which annotation should you use?

- A) @Controller
- B) @RestController
- C) @JsonController
- D) @WebController

Q24. What is the correct HTTP method to partially update a resource?

- A) PUT
- B) POST
- C) PATCH
- D) UPDATE

Q25. In Spring REST, which annotation binds a method parameter to a URI template variable (e.g., /users/{id})?

- A) @RequestParam
- B) @PathVariable
- C) @QueryParam
- D) @UriVariable

Q26. What HTTP status code typically represents 'Forbidden' (authenticated but lacks permissions)?

- A) 401 Unauthorized
- B) 403 Forbidden
- C) 404 Not Found
- D) 405 Method Not Allowed

Q27. Which interface does Spring Data JPA provide that includes pagination and sorting capabilities by default?

- A) CrudRepository
- B) PagingAndSortingRepository
- C) JpaPagination
- D) QuerydslPredicateExecutor

Q28. How can you externalize configuration for different environments (dev, prod) in Spring Boot?

- A) Using application-{profile}.properties and spring.profiles.active
- B) Creating multiple pom.xml files
- C) Using @EnvironmentConfig
- D) Spring Boot doesn't support environment profiles natively

Q29. Which class is the modern, non-blocking alternative to RestTemplate for making HTTP requests in Spring WebFlux?

- A) WebRequest
- B) HttpClient
- C) WebClient
- D) ReactiveRest

Q30. What annotation is used to globally handle exceptions across multiple @Controller classes in Spring Boot?

- A) @ExceptionHandler
- B) @ControllerAdvice
- C) @GlobalError
- D) @RestException

IV. React Framework

Q31. What is the primary purpose of the Virtual DOM in React?

- A) To completely replace the real DOM for security
- B) To improve performance by minimizing direct manipulation of the real DOM
- C) To allow React to be written in TypeScript
- D) To store application state globally

Q32. Which hook is used to access the Context API in a functional component?

- A) useContext
- B) useReducer
- C) useProvider
- D) useGlobalState

Q33. Why should you never mutate React state directly (e.g., state.count = 1)?

- A) It causes an infinite loop
- B) React will not know the state changed and will not re-render component

- C) It throws a compile-time error
- D) It deletes the Virtual DOM

Q34. In React Router (v6+), which component is used to define a mapping between a URL path and a Component?

- A) <Switch>
- B) <Route>
- C) <Link>
- D) <Navigate>

Q35. What is 'prop drilling' in React?

- A) Passing data from a child component to a parent
- B) Fetching data from a deeply nested API
- C) Passing props through multiple layers of components that don't need the data themselves
- D) Modifying a prop directly inside a component

Q36. What will happen if you forget to include a 'key' prop when rendering a list of elements via `Array.map()`?

- A) The code will fail to compile
- B) React will throw a runtime exception and crash the app
- C) React will log a warning in the console and may experience performance issues during re-renders
- D) Only the first element will be rendered

Q37. Which hook provides functionality similar to the 'componentDidMount' and 'componentDidUpdate' lifecycle methods?

- A) `useState`
- B) `useEffect`
- C) `useMemo`
- D) `useRef`

Q38. What does the `useRef` hook return?

- A) An array containing a stateful value and a setter function
- B) A mutable object whose `.current` property is initialized to the passed argument
- C) A memoized callback function
- D) A new Context object

Q39. In Redux, what is the role of a 'reducer'?

- A) To trigger asynchronous API calls
- B) To bind React components to the store
- C) To take the previous state and an action, and return the next state
- D) To define the UI structure

Q40. When optimizing a React app, what does `React.memo()` do?

- A) It caches API responses
- B) It memoizes a functional component, preventing re-renders if props haven't changed

- C) It stores variables in local storage
- D) It minifies the production build bundle

V. Git & Testing (JUnit/Mockito)

Q41. In Git, what command saves your uncommitted changes temporarily so you can switch branches?

- A) git pause
- B) git stash
- C) git save
- D) git cache

Q42. What is the effect of the 'git clone' command?

- A) It creates a duplicate branch locally
- B) It copies a remote repository to your local machine
- C) It copies a file within your repository
- D) It merges two branches

Q43. How do you rewrite the most recent commit message in Git?

- A) git commit --edit
- B) git rebase -msg
- C) git commit --amend
- D) git update-commit

Q44. What is a 'merge conflict' in Git?

- A) When two branches have different names
- B) When Git cannot automatically resolve differences in code between two commits being merged
- C) When a push is rejected by the remote server
- D) When a repository exceeds its storage limit

Q45. Which JUnit 5 annotation is used to execute a method before ALL tests in a class?

- A) @Before
- B) @BeforeEach
- C) @BeforeClass
- D) @BeforeAll

Q46. In Mockito, what is the default return value for a mocked method that returns an Object (and hasn't been stubbed)?

- A) An empty object
- B) null
- C) It throws a NullPointerException
- D) A new instance of the return type

Q47. What does the Mockito statement 'when(service.getData()).thenReturn("Mocked")' do?

- A) Verifies that getData() was called
- B) Defines a stub that returns 'Mocked' when getData() is invoked
- C) Injects 'Mocked' into the service dependency
- D) Mocks the entire service class automatically

Q48. When writing a unit test, why should you mock the database layer?

- A) To test the database performance
- B) To ensure the test only validates the business logic (Unit under test) independently
- C) Because databases cannot be used in testing environments
- D) To permanently modify the test database records

Q49. Which assertion method in JUnit 5 checks if a method throws a specific exception?

- A) assertThrows()
- B) assertException()
- C) expectThrows()
- D) @Test(expected=Exception.class)

Q50. What is the difference between @Mock and @InjectMocks in Mockito?

- A) @Mock injects dependencies, @InjectMocks creates them
- B) @Mock creates a mock object, @InjectMocks creates an instance of the class and injects the mocks into it
- C) They are functionally identical
- D) @InjectMocks is used for Spring Contexts only

Answer Key & Detailed Explanations

Q1: Answer B — Single Responsibility Principle (SRP) ensures a class only has one job or responsibility.

Q2: Answer A — BeanFactory is an implementation of the Factory design pattern, creating and managing beans.

Q3: Answer C — Spring beans are singletons by default, meaning one instance per Spring IoC container.

Q4: Answer C — The Decorator pattern attaches additional responsibilities to an object dynamically.

Q5: Answer A — Spring AOP uses dynamic proxies (JDK or CGLIB) to wrap target objects and execute cross-cutting concerns.

Q6: Answer C — Strategy pattern lets the algorithm vary independently from clients that use it.

Q7: Answer B — LSP states that objects of a superclass shall be replaceable with objects of its subclasses without breaking the application.

Q8: Answer C — The Observer pattern defines a one-to-many dependency so that when one object changes state, all its dependents are notified.

Q9: Answer A — Facade provides a higher-level, unified interface that makes the subsystem easier to use.

Q10: Answer B — DI is a technique to achieve Inversion of Control, where the framework manages object creation and binding.

- Q11: Answer B** — It occurs when lazy loading triggers an additional query for each child entity in a collection.
- Q12: Answer B** — `update()` transitions an object from detached to persistent state.
- Q13: Answer A** — `save()` immediately returns the generated ID, whereas `persist()` guarantees insertion but returns void.
- Q14: Answer B** — The 'compile' phase in the default lifecycle compiles the main source code.
- Q15: Answer C** — Test scope ensures the library (like JUnit) is not bundled in the production release.
- Q16: Answer A** — L1 cache is associated with the Session object and is enabled by default.
- Q17: Answer B** — `@IdClass` and `@EmbeddedId` are the standard JPA annotations for composite keys.
- Q18: Answer C** — `@Autowired` resolves dependencies by Type. (`@Resource` resolves by Name).
- Q19: Answer B** — HikariCP is the default, highly optimized connection pool in Spring Boot.
- Q20: Answer B** — Using `mappedBy` on the parent side and `@JoinColumn` on the child side establishes a bidirectional relationship without a join table.
- Q21: Answer B** — Actuator provides production-ready features to help you monitor and manage your application.
- Q22: Answer C** — `@EnableAutoConfiguration` triggers Spring Boot's auto-configuration mechanism.
- Q23: Answer B** — `@RestController` is a convenience annotation that combines `@Controller` and `@ResponseBody`.
- Q24: Answer C** — PATCH is used for partial modifications, while PUT is intended for complete resource replacement.
- Q25: Answer B** — `@PathVariable` extracts values from the URI path.
- Q26: Answer B** — 403 Forbidden indicates the server understood the request but refuses to authorize it.
- Q27: Answer B** — `PagingAndSortingRepository` extends `CrudRepository` to provide methods for pagination and sorting.
- Q28: Answer A** — Spring Profiles allow environment-specific configurations.
- Q29: Answer C** — `WebClient` is the reactive, non-blocking HTTP client introduced in Spring 5.
- Q30: Answer B** — `@ControllerAdvice` (or `@RestControllerAdvice`) allows you to consolidate exception handling logic globally.
- Q31: Answer B** — React computes changes in memory (Virtual DOM) and batches updates to the real DOM efficiently.
- Q32: Answer A** — `useContext` accepts a context object and returns the current context value.
- Q33: Answer B** — React relies on the `setState` (or `useState` setter) function to trigger the reconciliation and re-render process.
- Q34: Answer B** — The `<Route>` component defines the path and the element to render.
- Q35: Answer C** — Prop drilling happens when props are passed down deeply through intermediate components. Context API or Redux solves this.
- Q36: Answer C** — Keys help React identify which items have changed, are added, or are removed, optimizing list re-renders.

- Q37: Answer B** — useEffect allows you to perform side effects in functional components, running after renders.
- Q38: Answer B** — useRef provides a persistent, mutable reference that does not trigger re-renders when changed.
- Q39: Answer C** — Reducers are pure functions that determine how the application's state changes in response to an action.
- Q40: Answer B** — React.memo is a higher-order component that skips rendering if the component's props are identical to the previous render.
- Q41: Answer B** — git stash temporarily shelves changes you've made to your working copy.
- Q42: Answer B** — git clone downloads an existing remote repository, creating a local copy.
- Q43: Answer C** — git commit --amend allows you to modify the most recent commit.
- Q44: Answer B** — Conflicts occur when two branches have modified the same part of a file differently.
- Q45: Answer D** — @BeforeAll runs once before any of the test methods in the class (must be static). @BeforeClass was JUnit 4.
- Q46: Answer B** — By default, unstubbed methods returning objects in Mockito return null.
- Q47: Answer B** — This is standard Mockito stubbing, controlling the behavior of the mocked object.
- Q48: Answer B** — Unit tests must be isolated. Mocking the DB ensures you are only testing the logic of the service, not the DB configuration/connection.
- Q49: Answer A** — assertThrows() is the JUnit 5 standard for verifying exceptions. The @Test(expected=...) syntax is from JUnit 4.
- Q50: Answer B** — @Mock builds the fake object. @InjectMocks instantiates the real class you are testing and pushes the @Mocks into it.